

北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 2 - Student

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论文题目 Project Title	Network Resource Management Based on Language Model Multi-Agent Systems				
论文概述 Project outline Write about 500-800 words Please refer to Project Student Handbook section 3.2	Background and Motivation This project focuses on accelerating the communication pathway of KVCache transmission in large language model (LLM) training and inference, using the Huawei Ascend 910C NPU as the primary hardware platform [1]. In modern LLM systems—especially those deployed with high-performance distributed training or inference frameworks—the transfer of KVCache (Key–Value Cache) across devices has become a major performance bottleneck [2]. The challenge becomes more significant under the Prefill-Decode Separation (PD Separation) architecture. PD Separation decouples the model execution stages into Prefill (processing the input prompt and generating KVCache) and Decode (autoregressive token generation using the cached keys and values) [3]. This architecture improves parallelism and resource utilization but also increases the volume and frequency of KVCache transfers between system components [4]. Limitations of Existing Communication Mechanisms Traditional communication mechanisms such as RDMA (Remote Direct Memory Access) often struggle to meet the bandwidth and latency demands of KVCache transmission in PD-based systems. These limitations include insufficient effective bandwidth, fluctuations in latency, suboptimal path utilization, and scaling constraints when multiple workers are involved [5]. To overcome these challenges, this project aims to leverage SIO (Serial Input/Output)—a fundamental serial communication interface consisting of a one-to-one link between a master device and a slave device via a single data line and a clock line. The internal SIO interconnect of Ascend 910C offers high bandwidth and low latency, making it a promising alternative for high-speed KVCache transmission. Project Objective The key objective of the project is to design, implement, and integrate a new SIO-based communication module into the vLLM framework to accelerate KVCache transfer [2]. The target performance metric is to achieve at least 4× effective bandwidth improvement compared with the RDMA-based baseline in representative PD-separated scenarios. Requirement Analysis and Bottleneck Identification The project begins with an initial analysis of user/system requirements and identification of KVCache transmission bottlenecks. This stage includes profiling KVCache size patterns, understanding vLLM’s KV reuse and dispatching mechanisms, and evaluating the limitations of the RDMA communication path.				

	Data will be collected through Ascend 910C profiling tools, custom communication benchmarks, and vLLM execution traces [6]. Algorithms and Methodologies Next, various algorithms and methodologies will be explored. These include: - Analyzing SIO hardware characteristics (serial master-slave structure, clock-driven data synchronization, pipeline feasibility). - Designing optimized transmission strategies such as pipelined KV slicing, asynchronous forwarding, multilayer parallel scheduling, and adaptive buffer allocation [7]. - Building a plug-and-play communication abstraction that allows easy switching between RDMA mode and SIO mode within vLLM [8]. - Integrating the SIO communication backend into vLLM’s KVCache sharing modules (e.g., distributed KV reuse, remote KV transfer APIs) [9]. System Integration and User Interaction In terms of system interaction, users will be able to enable the SIO-accelerated transmission path via configuration or through an API interface in vLLM. The system should automatically detect hardware topology, initialize SIO channels, and manage transmission logic transparently. Experimental Design and Evaluation The experimental component will involve developing a reproducible benchmark suite capable of evaluating KVCache transmission under different conditions, such as varying KV dimensions, sequence lengths, numbers of layers, and numbers of decoding workers. Metrics including raw bandwidth, end-to-end latency, throughput, prefill-to-decode delay, and NPU/CPU utilization will be measured. These results will be compared directly against the RDMA-based baseline to verify whether the target speedup is achieved [10]. Tools, Languages, and Platform The project will use C++ and Python as the primary programming languages. Key tools include the Ascend CANN toolkit, AscendCL APIs, communication profiling utilities, and the vLLM execution engine. Background reference materials will include PD Separation literature, vLLM documentation, Ascend 910C communication manuals, and prior work on KVCache optimization and high-performance interconnects. References - Zuo P, Lin H, Deng J, et al. Serving Large Language Models on Huawei CloudMatrix384. arXiv preprint [Internet]. 2025 [cited 2025 Nov 17]; arXiv:2506.12708. Available from: https://arxiv.org/abs/2506.12708 - Kwon W, Li Z, Zhuang S, et al. Efficient Memory Management for Large Language Model Serving with PagedAttention (vLLM). arXiv preprint [Internet]. 2023 [cited 2025 Nov 17]; arXiv:2309.06180. Available from: https://arxiv.org/abs/2309.06180 - Zhong Y, Liu S, Chen J, et al. DistServe: Disaggregating Prefill and Decoding for Goodput-Optimized Large Language Model Serving. arXiv preprint [Internet]. 2024 [cited 2025 Nov 17]; arXiv:2401.09670. Available from: https://arxiv.org/abs/2401.09670 - Xiong Y, Wu H, Shao C, et al. LayerKV: Optimizing Large Language Model Serving with Layer-wise KV Cache Management. arXiv preprint [Internet]. 2024 [cited 2025 Nov 17]; arXiv:2410.00428. Available from: https://arxiv.org/abs/2410.00428
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	5. Li H, Li Y, Tian A, et al. A Survey on Large Language Model Acceleration Based on KV Cache Management. arXiv preprint [Internet]. 2024 [cited 2025 Nov 17]; arXiv:2412.19442. Available from: https://arxiv.org/abs/2412.19442 6. Cheng Y, Yang J, Zhang R, Liu F. LMCACHE: An Efficient KV Cache Layer for Enterprise-Scale LLM. arXiv preprint [Internet]. 2025 [cited 2025 Nov 17]; arXiv:2510.09665. Available from: https://arxiv.org/abs/2510.09665 7. Liao B, Zhang S, Wang Y, et al. Beyond KV Caching: Shared Attention for Efficient LLMs. arXiv preprint [Internet]. 2024 [cited 2025 Nov 17]; arXiv:2407.12866. Available from: https://arxiv.org/abs/2407.12866 8. Zhao J, Li J, Wu C. Sandwich: Separating Prefill-Decode Compilation for Efficient CPU LLM Serving. arXiv preprint [Internet]. 2025 [cited 2025 Nov 17]; arXiv:2507.18454. Available from: https://arxiv.org/abs/2507.18454 9. Yang J, Xu B, Li N, et al. KVLink: Accelerating Large Language Models via Efficient KV Cache Reuse. arXiv preprint [Internet]. 2025 [cited 2025 Nov 17]; arXiv:2502.16002. Available from: https://arxiv.org/abs/2502.16002 10. Du D, Cao S, Cheng J, Mai L, Cao T, Yang M. BitDecoding: Unlocking Tensor Cores for Long-Context LLMs with Low-Bit KV Cache. arXiv preprint [Internet]. 2025 [cited 2025 Nov 17]; arXiv:2503.18773. Available from: https://arxiv.org/abs/2503.18773
道德规范 Ethics Please discuss ethical issues with your supervisor. Please refer to Project Student Handbook section 4.1 and Project Ethics Guidance	Please confirm by checking the box: ☒ I confirm that I have discussed ethical issues with my supervisor. Summary of ethical issues: 1. Data Privacy and Security KVCache represents intermediate model states and may implicitly contain user-related semantic information. All experiments must use non-sensitive datasets, and secure data handling practices must be followed in accordance with university and Ascend platform requirements. 2. Algorithmic Transparency and Reproducibility The SIO-accelerated communication mechanism MUST NOT alter model semantics or change vLLM's inference outputs. All optimization logic, fallback mechanisms, and communication paths must be clearly documented and reproducible. 3. Fairness and Responsible Deployment Communication-layer optimization must not introduce unfair resource allocation (e.g., prioritizing certain requests) or system instability. The design must consider safe fallback to RDMA mode in case of failure. 4. Maintainability and Auditability The codebase should include proper documentation, tests, version tracking, and monitoring logs to enable debugging, auditing, and long-term maintenance. 5. Societal Impact and Sustainability Improvements in communication performance may enable larger-scale LLM deployments. The project should consider energy efficiency, resource consumption, and responsible scaling practices.

中期目标 Mid-term target. It must be tangible outcomes, E.g. software, hardware or simulation. It will be assessed at the mid-term oral.	By the mid-point of the project, the following tangible deliverables are expected: 1. SIO Communication Prototype A working SIO master-slave communication module on Ascend 910C. Successful basic data transfer demo validating SIO channel initialization and transmission. 2. KVCache Transmission Benchmark Suite Tools to simulate KVCache of different sizes and shapes (token length, number of layers, head dimensions). Supports both RDMA and SIO modes with automated performance measurement (bandwidth, latency, throughput). 3. Preliminary System Profiling Report Analysis of RDMA bottlenecks in PD Separation scenarios. Graphs showing bandwidth curves, prefill-to-decode pipeline timing, and resource usage. 4. Initial Thesis Draft Completed outline including Background, Problem Definition, Technical Approach, and Experiment Design.
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Work Plan (Gantt Chart)

Fill in the sub-tasks and insert a letter X in the cells to show the extent of each task

	Nov 1-15	Nov 16-30	Dec 1-15	Dec 16-31	Jan 1-15	Jan 16-31	Feb 1-15	Feb 16-28	Mar 1-15	Mar 16-31	Apr 1-15	Apr 16-30
Task 1 Requirement Analysis and Modeling Identify the key challenges in network resource management and model them as tasks suitable for language model multi-agent systems.												
Analyse PD-Separation (Prefill–Decode) workflow and KVCache transmission patterns	X	X										
Benchmark RDMA-based KVCache transmission & identify bottlenecks	X	X	X									
Study Ascend 910C SIO hardware characteristics and communication constraints		X	X	X								
Review vLLM’s KVCache reuse/dispatching architecture			X	X								
Task 2 Design and Development of Multi-Agent Framework Build a framework where language model agents can interpret network requirements, interact, and coordinate resource allocation decisions.												
Design SIO-based KVCache transmission architecture					X	X						
Define data slicing / pipelining mechanism for high-throughput KV transfer						X	X					
Develop communication abstraction layer (switchable RDMA ↔ SIO)							X	X				
Integrate SIO backend with vLLM’s KVCache transport interface								X	X			
Task 3 Implementation of Algorithms and Tools Apply optimization techniques, reinforcement learning, and semantic reasoning within the multi-agent system using tools such as Python, PyTorch, and network simulators.												
Code SIO master-slave communication module								X	X			
Implement KVCache sender / receiver pipelines									X	X		
Develop profiling suite (BW / latency)										X	X	
Optimize buffers / parallelism / async scheduling											X	X
Task 4 Evaluation and Performance Validation Test the framework in simulated network environments (e.g., ns-3) to measure scalability, efficiency, and adaptability, and compare results against traditional approaches.												
Prepare experimental bed (Ascend 910C + vLLM)										X	X	

Run PD pipeline latency tests											X	X
Compare RDMA vs SIO throughput & Bandwidth												X
Analyse results & write performance report												X